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Relatively long plateau → lasts about 175 ms

membrane potential reaches approx zero the Ca^{2+} channels close & K^+ channels allowing K^+ to exit the cell.

The repolarization lasts about 75 ms

→ @ this point, membrane potential drops until it reaches levels once more & the cycle repeats. The entire event lasts between 250 & 300 ms.

absolute -> refactory period for cardiac contractile muscle lasts about 200 ms

Refactory period lasts → approx 50 ms, since extended period is critical the heart muscle must contract, to total 250 ms pump blood effectively & the contraction must follow the electrical events. Without extended refactory periods → premature contractions would occur in the heart & will not be compatible with life.





